

① Laminar & Turbulent flow:

Streamline flow.

Raynold's No.

$$N_R = \frac{\rho v D}{\eta}$$

density ρ →
 vel. v →
 cross-section dia D →
 Coeff of Viscosity η →

Streamlines

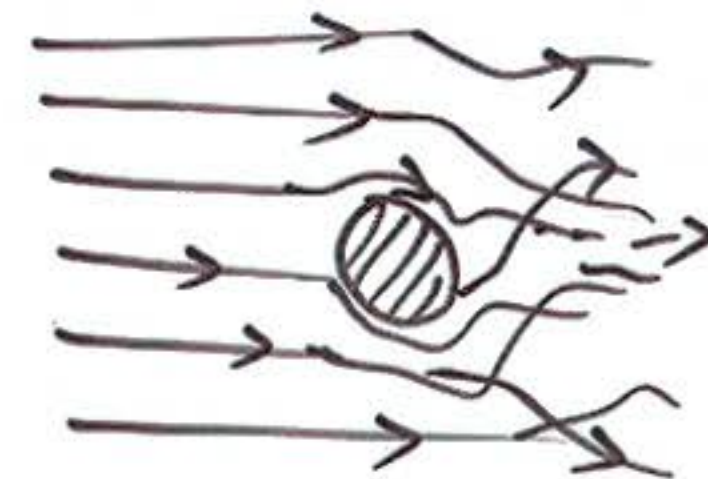


A diagram of a vertical tube with a stopper at the top. Arrows labeled 'Streamlines' point downwards inside the tube, indicating laminar flow. A label 'v' with a downward arrow is next to the tube.

Generally if

$$N_R > 2000$$

then flow becomes turbulent



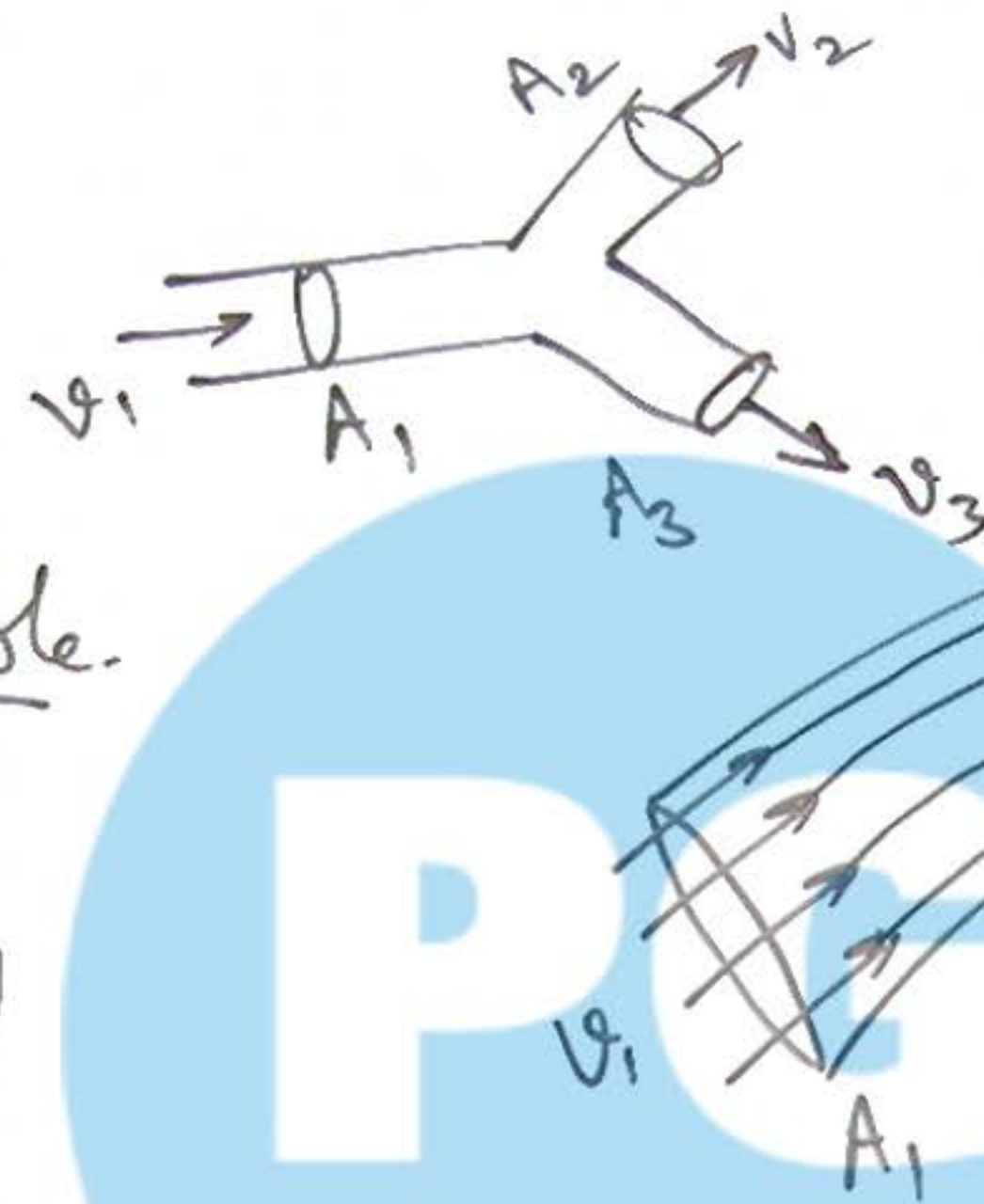
② Equation of Continuity ✓

fluid $\rightarrow$ incompressible.

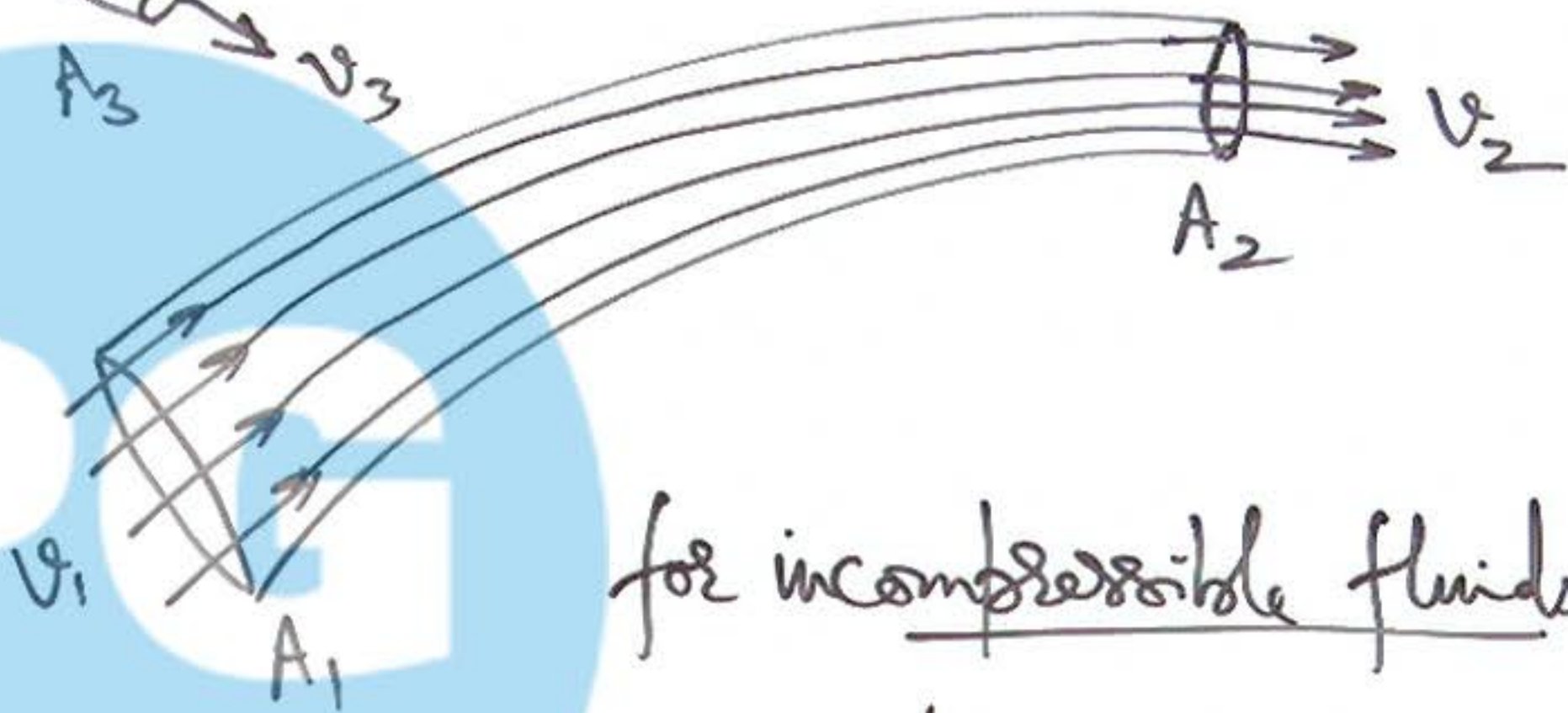
for any flowing fluid

Volume flow rate

$$Q = \frac{dV}{dt} = Av$$



$$A_1 v_1 = A_2 v_2 + A_3 v_3$$



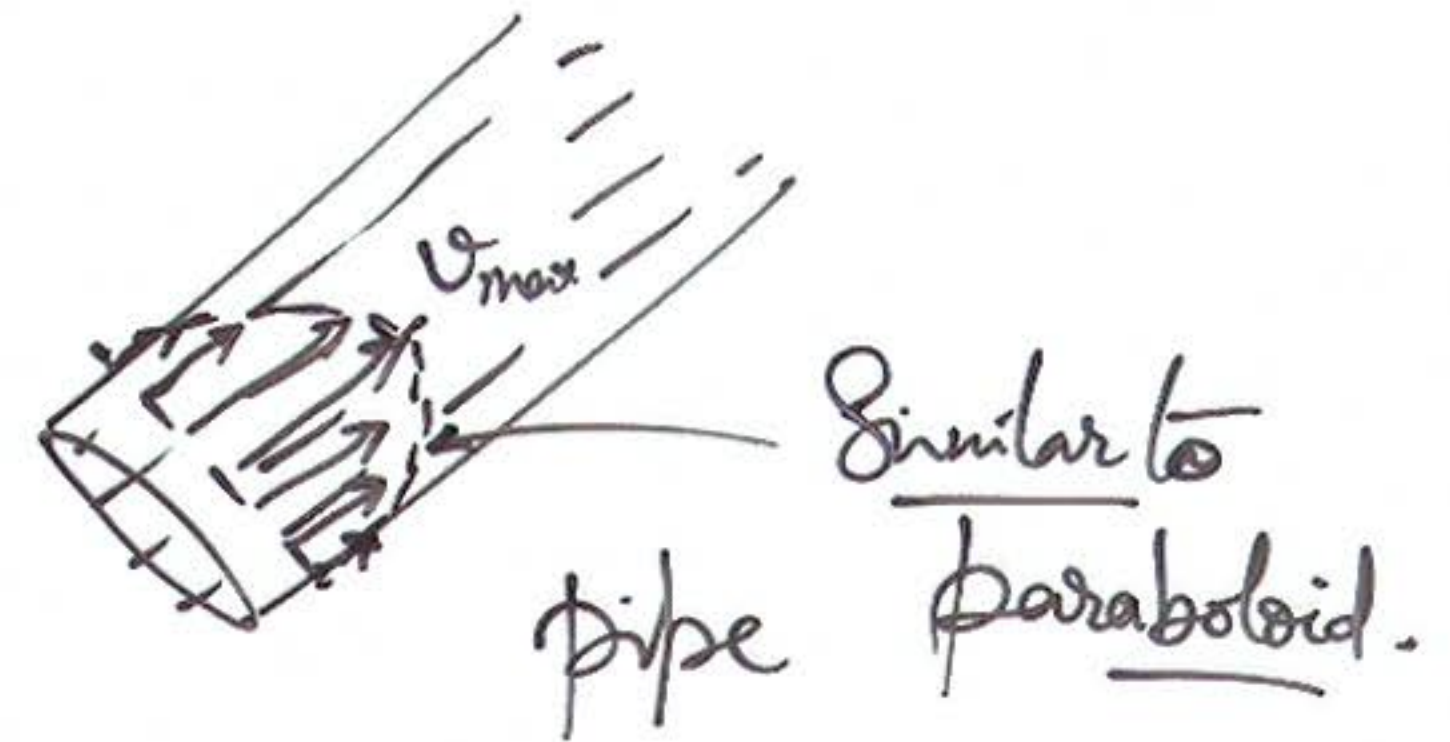
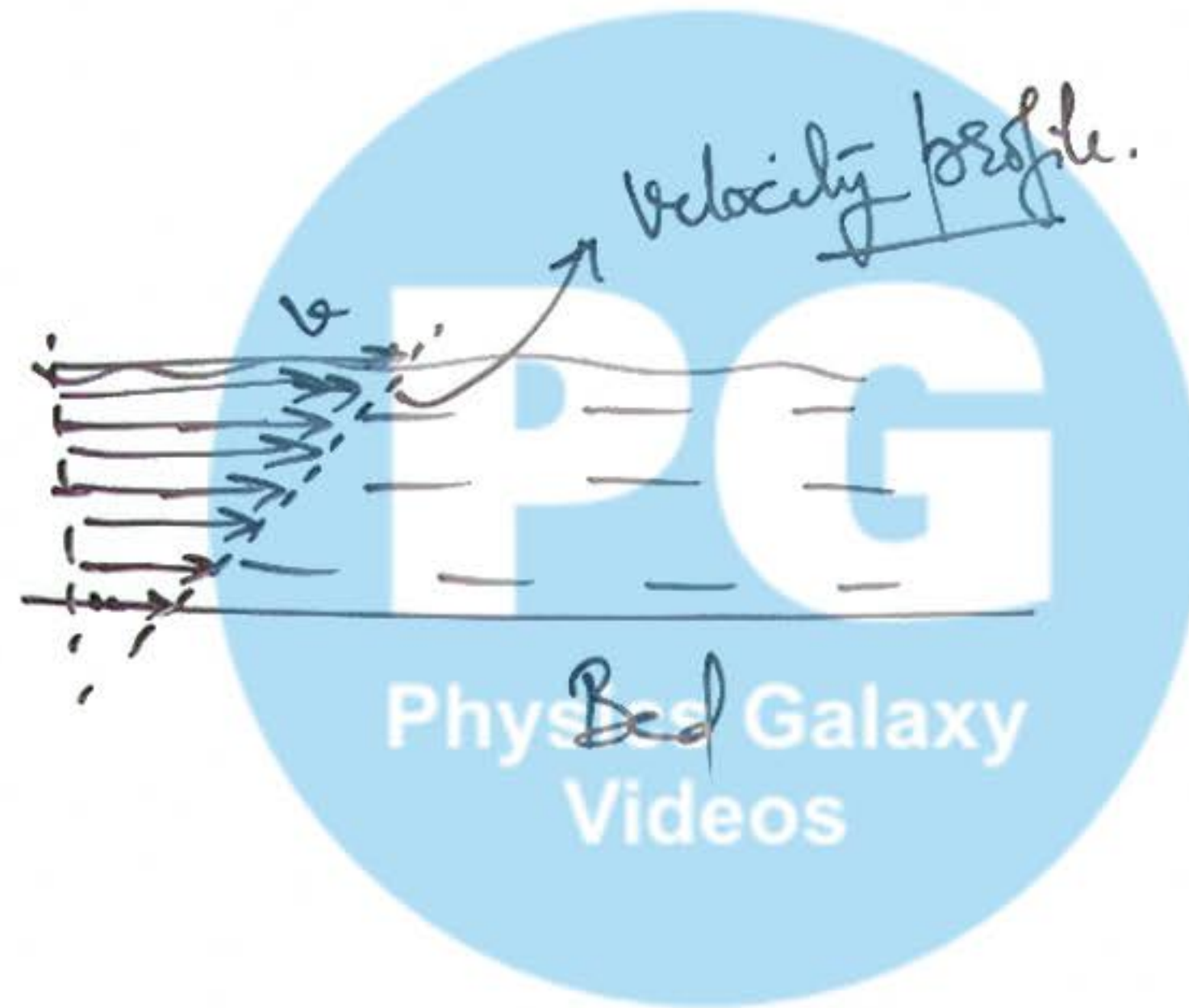
for incompressible fluids

we always use

$$A_1 v_1 = A_2 v_2 \quad \checkmark$$

$$Av = \text{Constant}$$

③ Velocity Profile of a
flowing fluid:



④ Bernoulli's Theorem:

Energy cons for
flowing fluid.

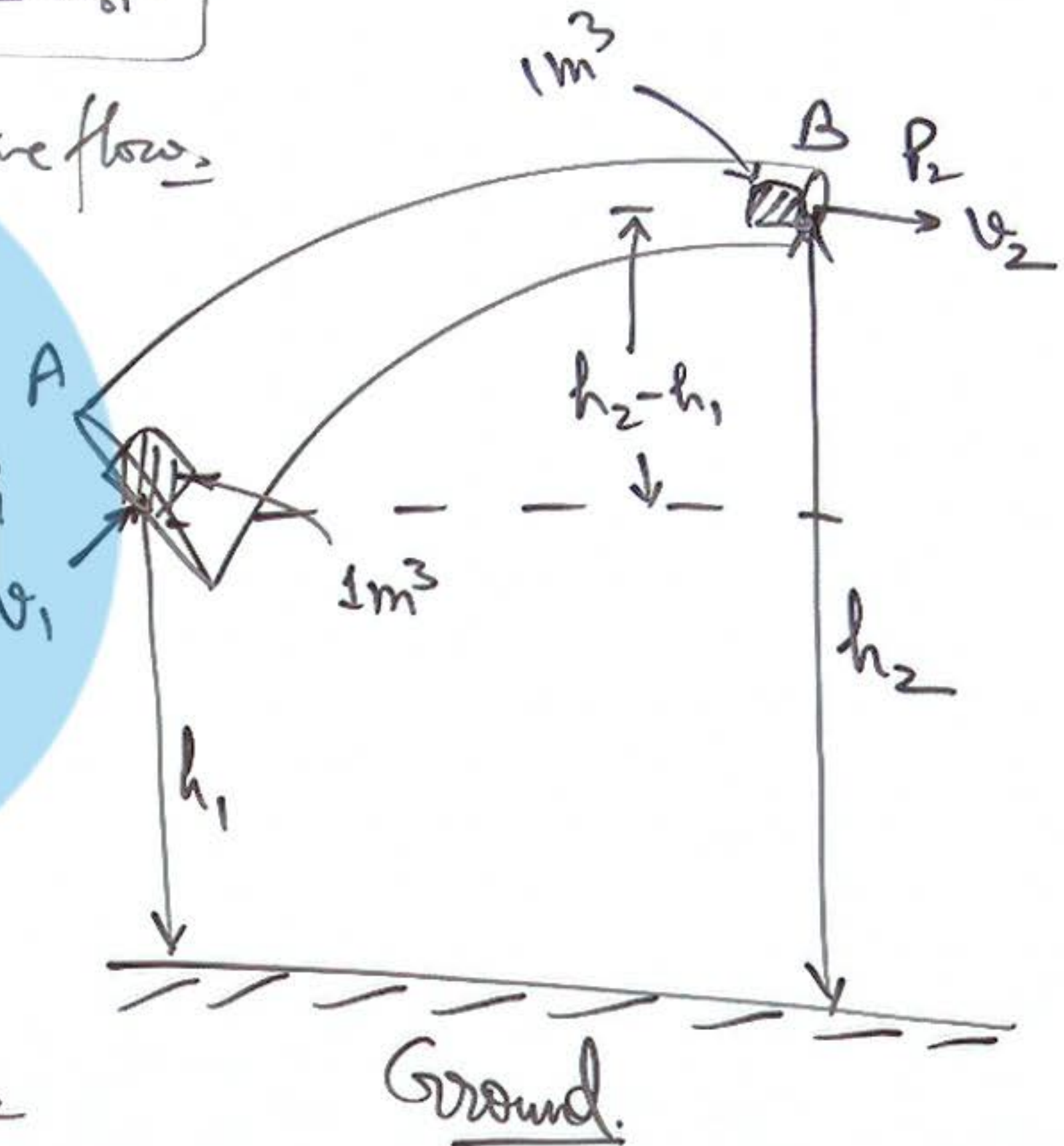
W/E theorem for 1 m^3 vol flowing from A to B.

$$\frac{1}{2}\rho v_1^2 + P_1 - \rho g(h_2 - h_1) - P_2(1) = \frac{1}{2}\rho v_2^2$$

$$\frac{1}{2}\rho v_1^2 + P_1 + \rho g h_1 = \frac{1}{2}\rho v_2^2 + \rho g h_2 + P_2$$

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{Const}$$

Valid only for streamline flow.



⑤ P-v Trade off:

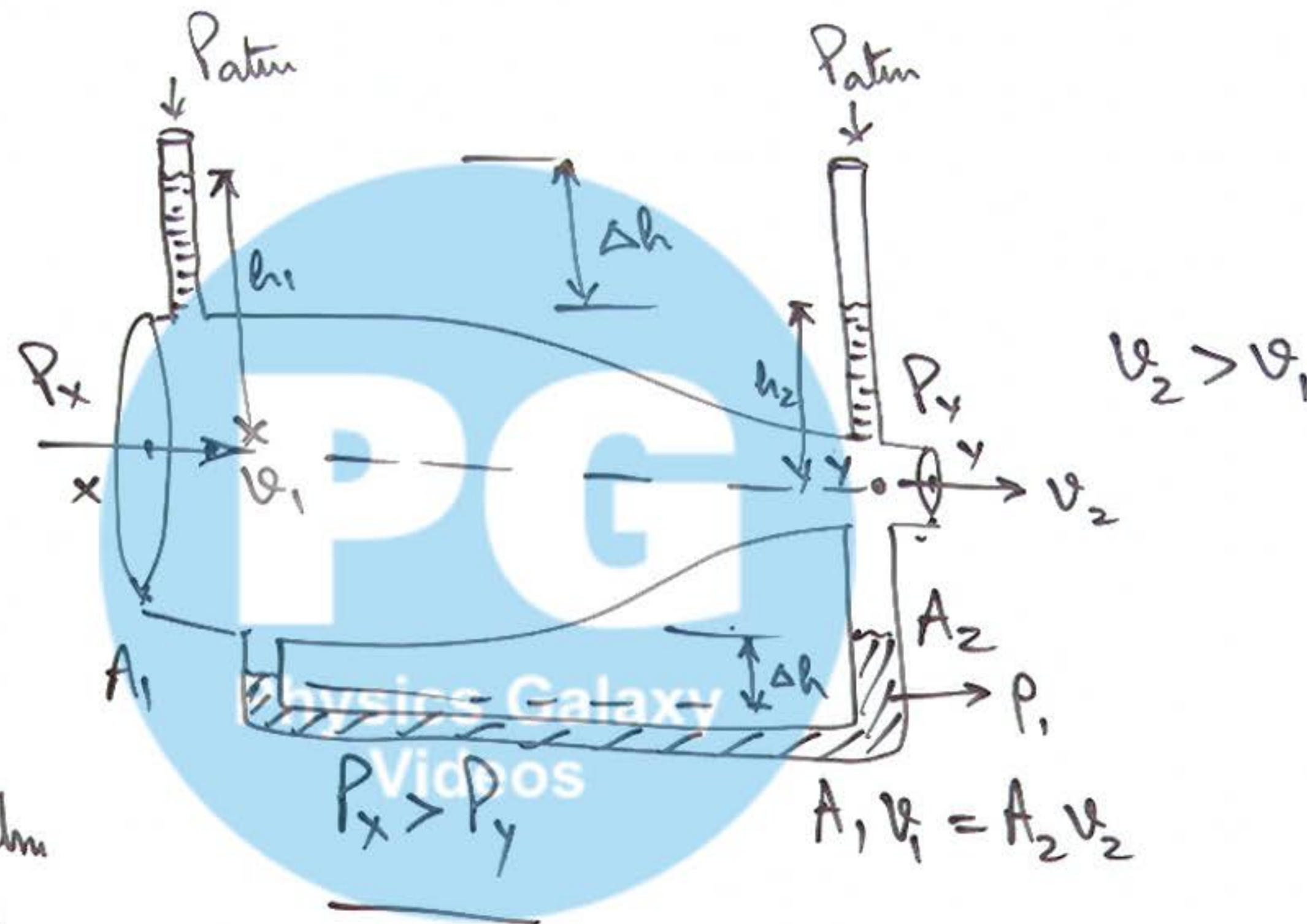
By B.T.

$$\checkmark \underline{P} + \frac{1}{2} \rho \underline{v}^2 = \text{Constant}$$

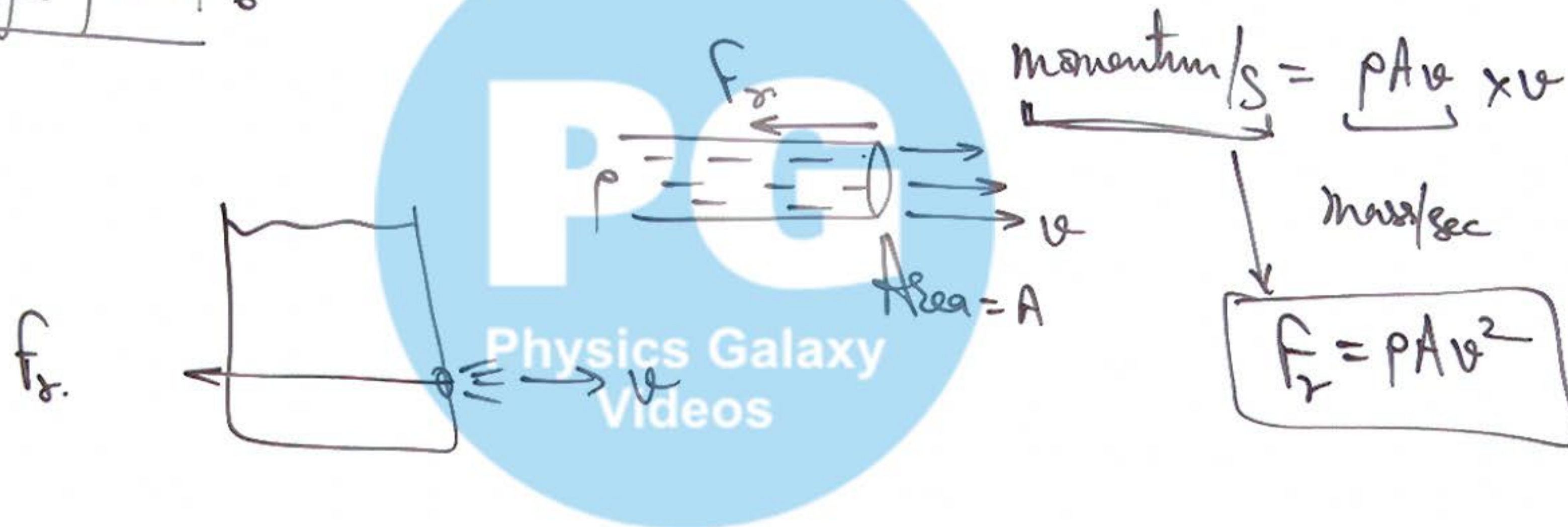
$$P_x = h_1 \rho g + P_{\text{atm}}$$

$$P_y = h_2 \rho g + P_{\text{atm}}$$

$$P_x - P_y = \rho g \Delta h$$



(5A) Reaction force due to ejection of fluid:



⑥ Torricelli's Theorem:

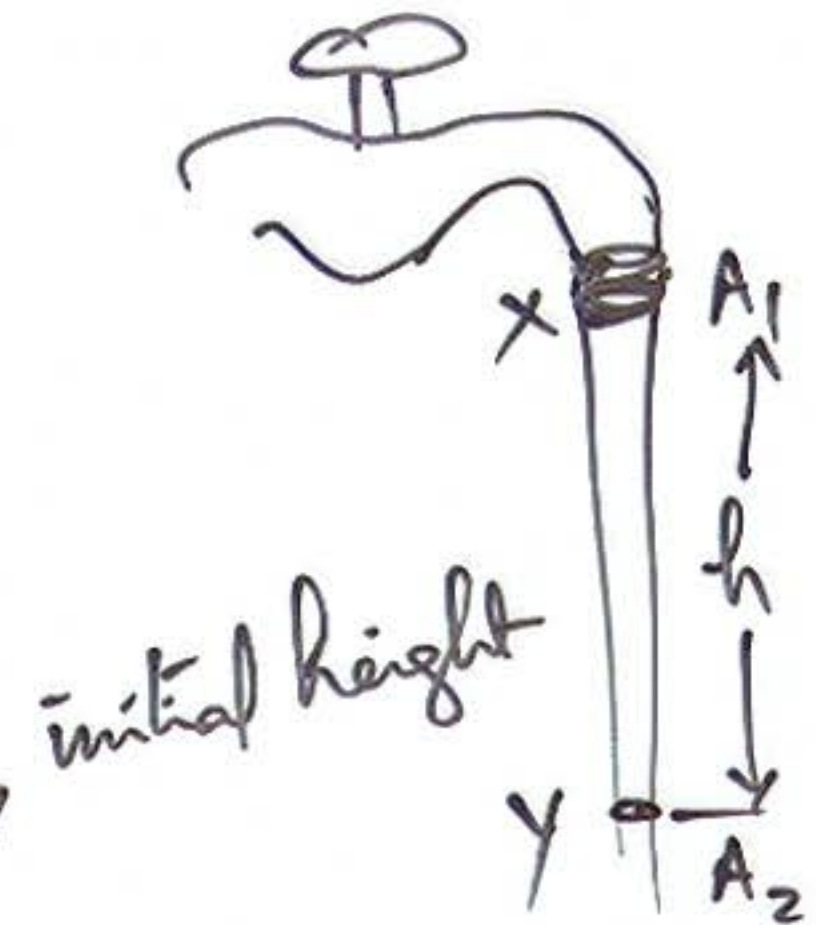
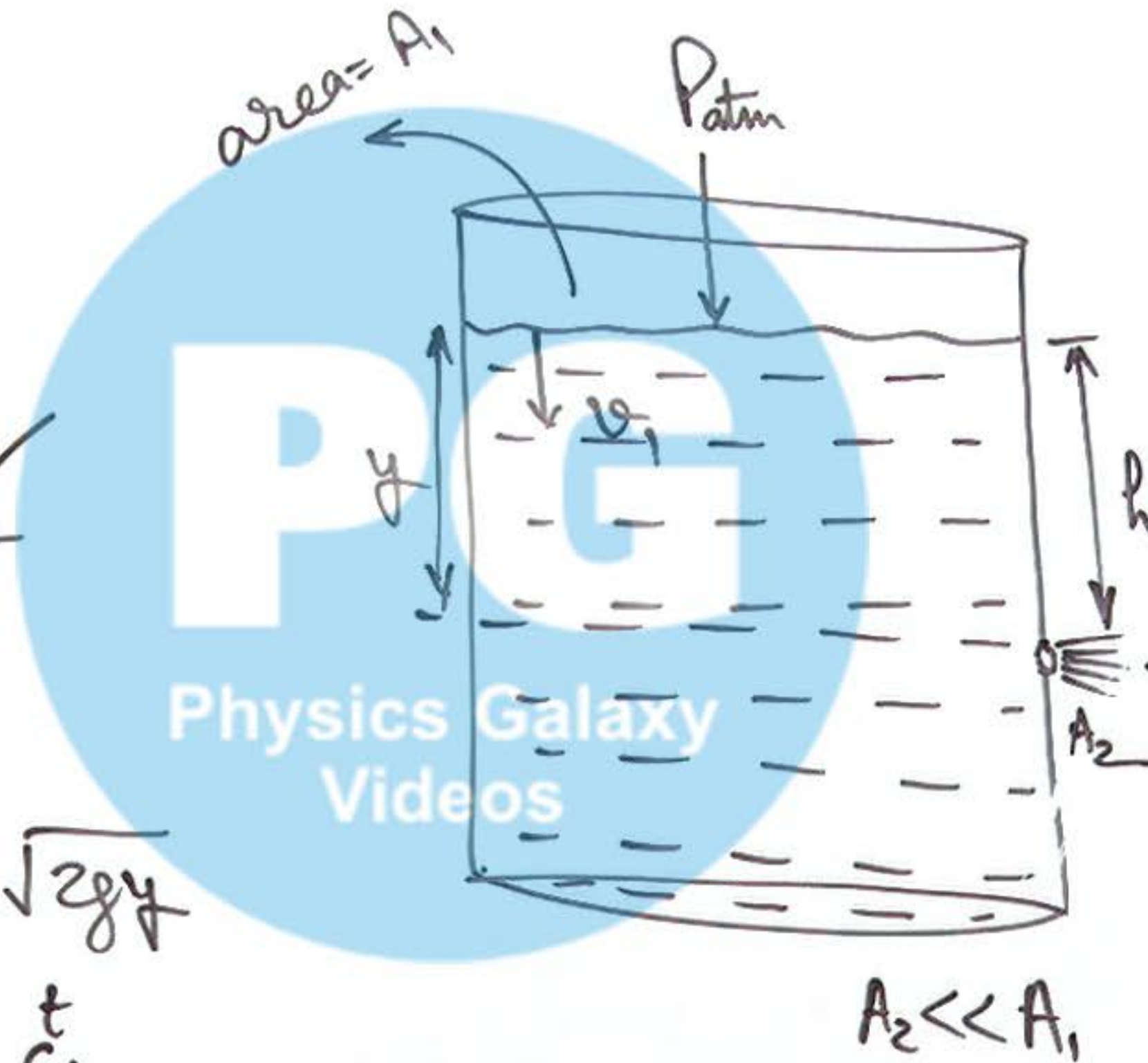
By Eqⁿ of Cont.

$$A_1 v_1 = A_2 v_2$$

for small area $A_2 \ll A_1$

$$-\frac{dy}{dt} = v_1 = \frac{A_2}{A_1} \sqrt{2gy}$$

$$\int_h^0 \frac{dy}{\sqrt{y}} = -\int_0^t \frac{A_2}{A_1} \sqrt{2g} dt \rightarrow 2\sqrt{h} = \frac{A_2}{A_1} \sqrt{2g} \cdot t \rightarrow t = \frac{2\sqrt{h} A_1}{A_2 \sqrt{2g}}$$



$$v_2 = \sqrt{2gh}$$

$$v_2 = \sqrt{\frac{2gh}{1 - A_2^2/A_1^2}}$$